CASE STUDY — TASK

Inventory Management System

Java | Object-Oriented Programming | Console Application

Language	Java	Application	Console-based
Core classes	Inventory, InventoryManagement	Persistence	Not required

1. Objective

Apply Object-Oriented Programming (OOP) principles to design and implement simple, modular Java classes for an inventory management application.

2. Problem Statement

Create a Java program that allows users to manage an inventory by adding items with their respective quantities, removing items from the inventory, and checking the quantity of items in the inventory. After one transaction is completed, the user must be able to choose another action until the user chooses to exit.

3. Mandatory Class Requirements

Class	Responsibility
Inventory	Manage inventory data and provide operations to add items, remove items, and check item quantities.
InventoryManagement	Provide the console user interface, display the menu, accept input, and call methods of Inventory.

4. Required Menu

Option	Action	Expected behaviour
1	Add Item	Accept an item name and quantity. Add the quantity to the inventory. If the item already exists, increase its ex
2	Remove Item	Accept an item name and quantity to remove. Do not allow removal if the item does not exist or the available
3	Check Quantity	Accept an item name and display its current quantity, or an appropriate message if it is not present.
4	Exit	Terminate the application.

5. Functional Requirements

• The menu must be displayed repeatedly until the user selects Exit.

• Item names must not be blank and quantities must be positive integers.

• Adding an existing item must increase its current quantity rather than create a duplicate entry.

• Removing an item must reduce its quantity. If its quantity becomes zero, remove the item from the inventory.

• The program must not allow removal of more than the available quantity.

• Checking an item must display its current quantity.

• Display suitable messages for missing items, invalid input, and invalid menu choices.

• Inventory data only needs to remain available while the program is running. Saving data to a file or database is not required.

6. Suggested Design

Use an Inventory object to hold inventory data. A Map such as HashMap<String, Integer> is a suitable implementation: the item name is the key and its quantity is the value. Keep inventory operations inside the Inventory class. Keep input, output, and menu control inside InventoryManagement.

7. Expected Deliverables

• Inventory.java containing inventory data and business operations.

• InventoryManagement.java containing the console menu and user interaction.

• A working demonstration of adding, removing, and checking item quantities.

• No file or database storage is required.

Learning outcome: Demonstrate encapsulation, separation of responsibilities, object creation, and method interaction between modular Java classes.